

Insurance Cost Analysis

```
insurance <- read.csv("data/medical_insurance.csv")
names(insurance)
```

```
[1] "person_id"           "age"
[3] "sex"                 "region"
[5] "urban_rural"         "income"
[7] "education"           "marital_status"
[9] "employment_status"   "household_size"
[11] "dependents"          "bmi"
[13] "smoker"              "alcohol_freq"
[15] "visits_last_year"    "hospitalizations_last_3yrs"
[17] "days_hospitalized_last_3yrs" "medication_count"
[19] "systolic_bp"         "diastolic_bp"
[21] "ldl"                 "hba1c"
[23] "plan_type"           "network_tier"
[25] "deductible"          "copay"
[27] "policy_term_years"   "policy_changes_last_2yrs"
[29] "provider_quality"    "risk_score"
[31] "annual_medical_cost" "annual_premium"
[33] "monthly_premium"     "claims_count"
[35] "avg_claim_amount"    "total_claims_paid"
[37] "chronic_count"       "hypertension"
[39] "diabetes"            "asthma"
[41] "copd"               "cardiovascular_disease"
[43] "cancer_history"      "kidney_disease"
[45] "liver_disease"       "arthritis"
[47] "mental_health"       "proc_imaging_count"
[49] "proc_surgery_count"  "proc_physio_count"
[51] "proc_consult_count"  "proc_lab_count"
[53] "is_high_risk"        "had_major_procedure"
```

Introduction

In the United States, health insurance costs rank amongst the highest in the world. Within the year 2023, healthcare spending increased by 7.5%, totaling up to 4.9 trillion USD (ama). This consequently drives up the price of health insurance premiums. Recently, the rate of change in health insurance premiums has exceeded that of wages and inflation, making it harder for individuals to afford health insurance. However, health insurance costs vary from one individual to another; factors such age, location, and tobacco use are legally allowed increase the cost of one's premium. Further, individuals with other pre-existing health risks may themselves select costlier healthcare plans that provide greater coverage.

Consequently, various factors may contribute to the cost of one's premium. However, these factors may be split into two groups: modifiable lifestyle choices and baseline demographic/socioeconomic factors. This study seeks to figure out what proportion of insurance cost variance is attributable to modifiable lifestyle choices versus baseline demographic and socioeconomic risk factors. Being able to see these results can inform the public on the factors they should be paying attention to if they want to lower their premiums. For example, if insurance cost variance is heavily driven by modifiable lifestyle choices such as smoking or alcohol consumption, prices are determined heavily on the behaviors of the customer. However, if the insurance cost variance is more controlled by the socioeconomic factors, then there must be a change in the system in order to lower costs.

For the specific variables considered, within modifiable lifestyle choices, there is smoking status (current, former, or never), frequency of alcohol consumption (Never, occasionally, weekly, daily), and BMI. As for the baseline demographic/socioeconomic factors category, it contains age, sex, highest education level, income, marital status, and household size. The dataset containing all of these variables was obtained from Kaggle; the Kaggle dataset got it's data from 3 main sources: Centers for Disease Control and Prevention (CDC), U.S. Department of Health & Human Services (HHS), and Centers for Medicare & Medicaid Services (CMS). In the scope of our research question, the dataset contained an excess amount of information; thus, those columns were filtered out and we only looked at the variables apart of the two main categories. Throughout the paper, we will run regressions and visualize the data in order to see how the two categories influence the variance of medical insurance costs.

Research Question: What proportion of insurance cost variance is attributable to modifiable lifestyle choices versus baseline demographic and socioeconomic risk factors?

Methodology

As we sought to investigate the contributing effect of a variety of variables on an output, monthly insurance premium, the use of a linear model (either `lm` or `glm`) was suitable for the situation. We conducted initial tests to evaluate whether a linear model (`lm`) or generalized linear model (`glm`) was appropriate. Fitting a linear model to variables representing modifiable

choices and another to those representing demographics, we created residual plots and residual histograms for each linear model. Both residual histograms created showed significant right skew, suggesting that errors were not normally distributed - an assumption necessary for the use of a linear model. Consequently, we utilized instead a generalized linear model (glm) in the following steps.

After fitting a generalized linear model, we conduct bivariate tests (why so?) for variables representing modifiable factors. As alcohol consumption and tobacco use are categorical variables, we utilize an ANOVA test. To do so, however, we also generate graphs to verify that the assumptions for ANOVA are satisfied. For both cases, we can assume that the outcomes are independent, and that each “subject” is not affecting another. CHECK REQUIREMENTS FOR ANOVA WHEN CATEGORICAL VARIABLES ARE UTILIZED.

Results

Initial bivariate analyses revealed differential relationships between modifiable lifestyle factors and annual insurance premiums. BMI demonstrated a statistically significant but weak positive correlation with premium costs ($r = 0.042$, 95% CI [0.035, 0.048], $p < 0.001$), suggesting that BMI alone explains minimal variance in insurance costs. In contrast, smoking status showed a substantial and highly significant association with premiums ($F(2, 99997) = 1247$, $p < 0.001$), indicating meaningful differences in insurance costs across smoking categories. Notably, alcohol consumption frequency was not significantly associated with premium costs ($F(3, 99996) = 0.597$, $p = 0.617$), suggesting that alcohol use patterns do not independently predict insurance premiums in this sample. These preliminary findings suggest that while smoking status is a strong univariate predictor of insurance costs, the relationship between BMI and premiums may be more complex and potentially moderated by other factors, warranting multivariate analysis to account for confounding variables and interactions.

Figure [X] displays the distribution of annual insurance premiums across smoking status categories. The violin plots reveal substantial differences in premium distributions among the three groups. Current smokers demonstrate notably elevated premiums with a median around 1,200 USD and considerable variability extending to over 1,500 USD for the upper quartile. Former smokers show an intermediate distribution, with lower median premiums than current smokers but higher than never-smokers. Never-smokers exhibit the lowest and most concentrated premium distribution, with the majority of observations clustering below 1,000 USD. The pronounced vertical separation between groups—particularly between current smokers and never-smokers—provides preliminary visual evidence that smoking status, a modifiable lifestyle factor, is strongly associated with insurance costs. The overlapping distributions suggest some shared variance with other factors, which will be explored through formal variance partitioning analysis.

Model 1 explained 2.6% of premium variance and was statistically significant ($F = 448.3$, $p < 0.001$). Smoking status was the strongest predictor. Compared to current smokers, former

smokers paid \$141 less annually and never smokers paid \$192 less (both $p < 0.001$), representing roughly 20 to 26% lower premiums. BMI showed a positive association, with each unit increase linked to a \$3.43 premium increase ($p < 0.001$). Alcohol consumption was not significantly related to premiums at any level (all $p > 0.15$).

Model 2 explained only 1.6% of premium variance. Age was the strongest baseline predictor, with each year associated with a \$3.19 premium increase ($p < 0.001$). Income showed a small negative effect ($p = 0.032$), while sex, marital status, and household size were not significant.

Model 3 combined all predictors and explained 4.3% of variance, nearly the sum of Models 1 and 2. Key predictors maintained stable effect sizes. Smoking coefficients were nearly identical to Model 1 (former: \$141 less, never: \$194 less, both $p < 0.001$), as were BMI and age effects. This stability suggests minimal confounding between modifiable and baseline factors.

Model 4 tested for BMI by smoking interactions but explained virtually the same variance as Model 3 (4.29% vs 4.27%). The interaction terms were not significant ($p = 0.42$ and $p = 0.50$), indicating the relationship between BMI and premiums does not differ across smoking groups. We therefore selected Model 3 as our final model.

The low R-squared values across all models (1.6% to 4.3%) indicate that most premium variance is unexplained by measured predictors. This likely reflects unmeasured factors such as coverage type, deductibles, policy terms, or company-specific pricing. However, the significant effects of modifiable factors, particularly smoking status, demonstrate that lifestyle choices contribute meaningfully to premium determination.

```
#Ran a regression for modifiable life choices; code has been switched to glm as assumptions a
mM <- lm(annual_premium ~ bmi + smoker + alcohol_freq, data=insurance)
summary(mM)
```

Call:

```
lm(formula = annual_premium ~ bmi + smoker + alcohol_freq, data = insurance)
```

Residuals:

Min	1Q	Median	3Q	Max
-536.0	-220.4	-112.2	83.1	10204.1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	655.3464	9.3629	69.994	<2e-16 ***
bmi	3.4341	0.2497	13.754	<2e-16 ***
smokerFormer	-141.0053	4.6245	-30.491	<2e-16 ***
smokerNever	-192.4150	3.8799	-49.593	<2e-16 ***
alcohol_freqNone	-5.2296	6.0193	-0.869	0.385

alcohol_freqOccasional	-5.8825	5.8747	-1.001	0.317
alcohol_freqWeekly	-8.8204	6.2373	-1.414	0.157

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 394.3 on 99993 degrees of freedom

Multiple R-squared: 0.02619, Adjusted R-squared: 0.02614

F-statistic: 448.3 on 6 and 99993 DF, p-value: < 2.2e-16

```
mB <- lm(annual_premium ~ age + sex + education + income + marital_status + household_size,
summary(mB))
```

Call:

```
lm(formula = annual_premium ~ age + sex + education + income +
    marital_status + household_size, data = insurance)
```

Residuals:

Min	1Q	Median	3Q	Max
-488.2	-223.2	-114.1	83.2	10245.2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.268e+02	7.089e+00	60.210	<2e-16	***
age	3.186e+00	7.839e-02	40.638	<2e-16	***
sexMale	2.538e+00	2.532e+00	1.002	0.3162	
sexOther	-1.051e+00	9.012e+00	-0.117	0.9072	
educationDoctorate	-7.176e+00	7.663e+00	-0.936	0.3490	
educationHS	3.658e+00	3.455e+00	1.059	0.2897	
educationMasters	7.065e+00	4.104e+00	1.722	0.0851	.
educationNo HS	1.285e+01	6.024e+00	2.132	0.0330	*
educationSome College	3.972e+00	3.445e+00	1.153	0.2489	
income	-5.740e-05	2.678e-05	-2.143	0.0321	*
marital_statusMarried	-1.665e+00	5.213e+00	-0.319	0.7494	
marital_statusSingle	-1.915e+00	5.185e+00	-0.369	0.7119	
marital_statusWidowed	-3.340e+00	7.829e+00	-0.427	0.6697	
household_size	1.690e+00	1.318e+00	1.282	0.1998	

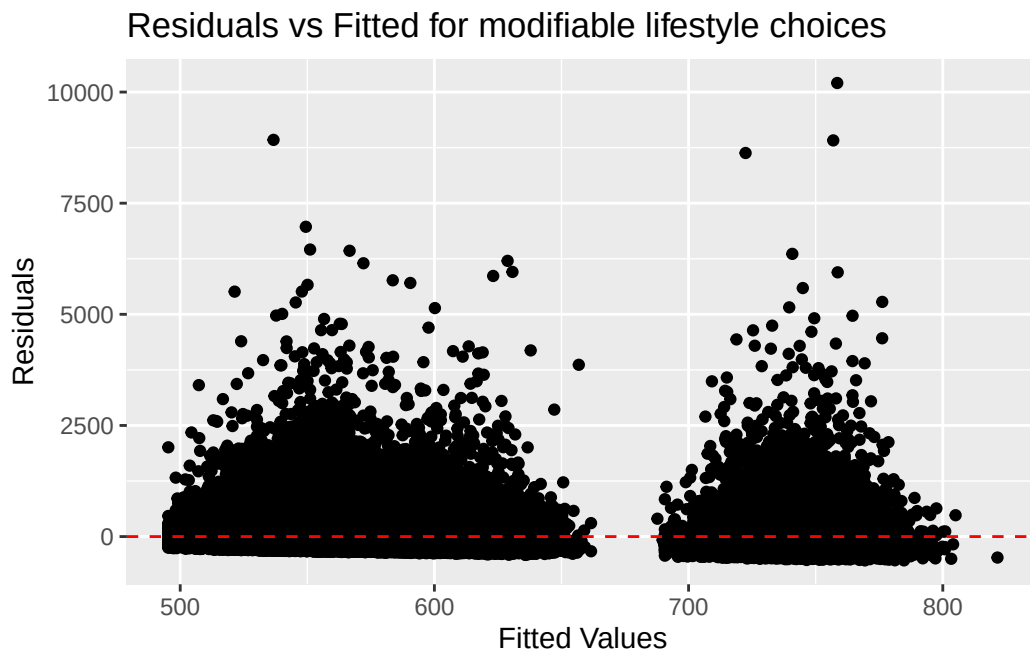
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 396.3 on 99986 degrees of freedom

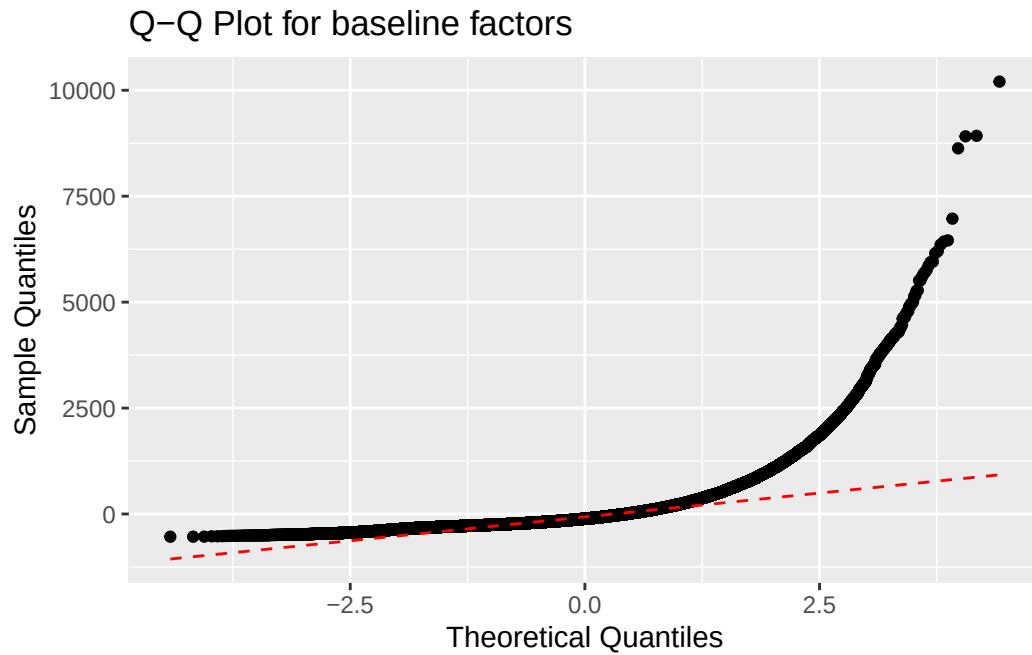
Multiple R-squared: 0.01641, Adjusted R-squared: 0.01628

F-statistic: 128.3 on 13 and 99986 DF, p-value: < 2.2e-16

```
#Checking assumptions for the models
ggplot(data.frame(fitted = fitted(mM), residuals = residuals(mM)),
       aes(x = fitted, y = residuals)) +
  geom_point(color = "black") +
  geom_hline(yintercept = 0, color = "red", linetype = "dashed") +
  labs(title = "Residuals vs Fitted for modifiable lifestyle choices",
       x = "Fitted Values", y = "Residuals")
```

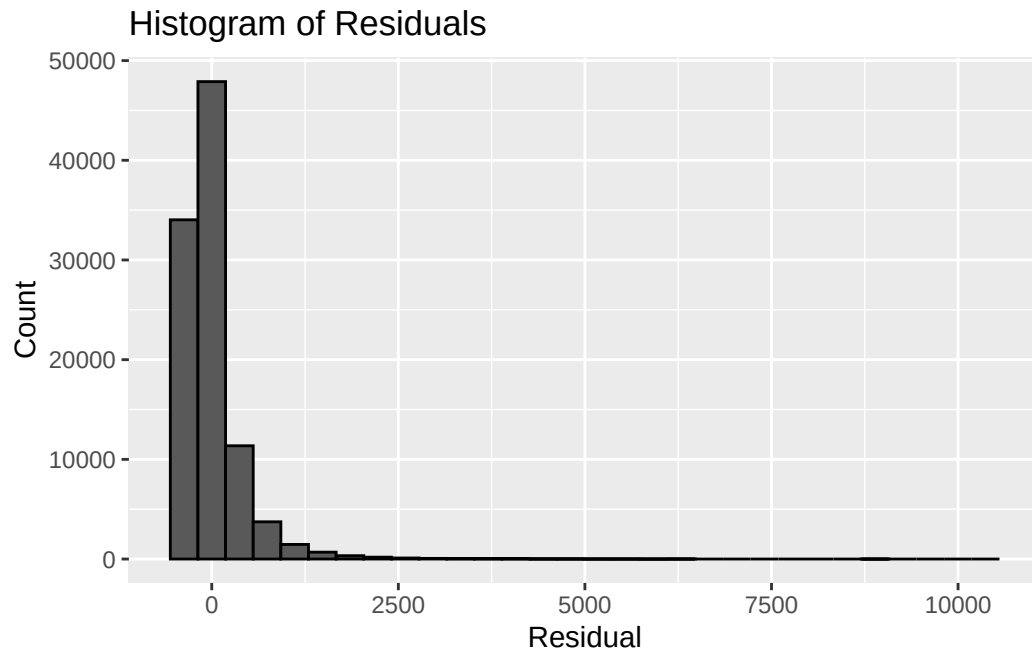


```
ggplot(data.frame(sample = residuals(mM)), aes(sample = sample)) +
  stat_qq(color = "black") +
  stat_qq_line(color = "red", linetype = "dashed") +
  labs(title = "Q-Q Plot for baseline factors",
       x = "Theoretical Quantiles", y = "Sample Quantiles")
```

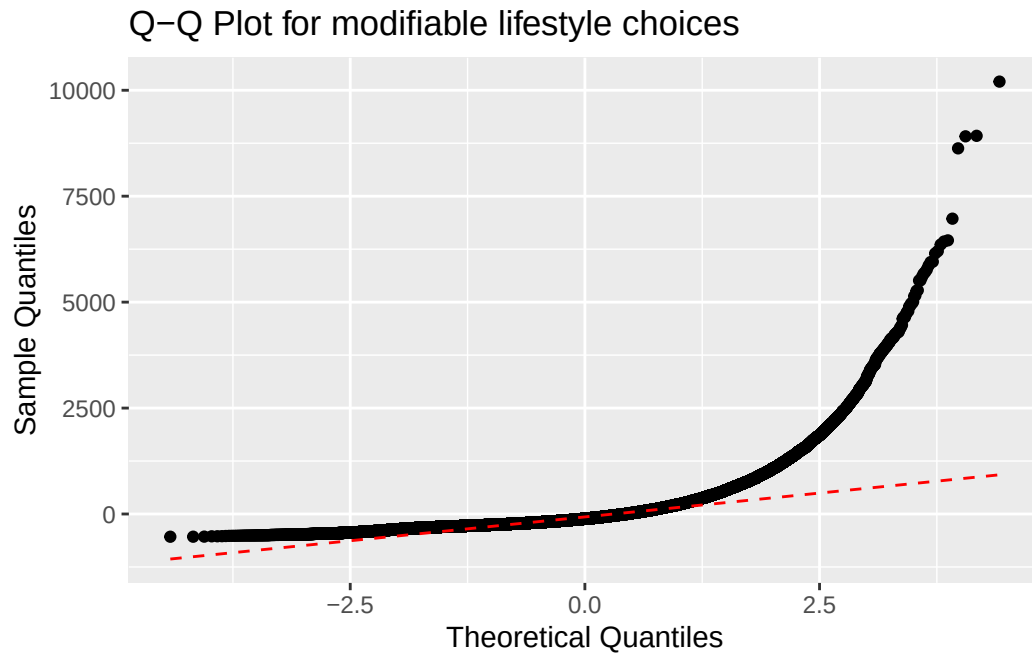


```
ggplot(data = mM, mapping = aes(x = mM$residuals)) +  
  geom_histogram(color = "black") +  
  labs(title = "Histogram of Residuals",  
        x = "Residual", y = "Count")
```

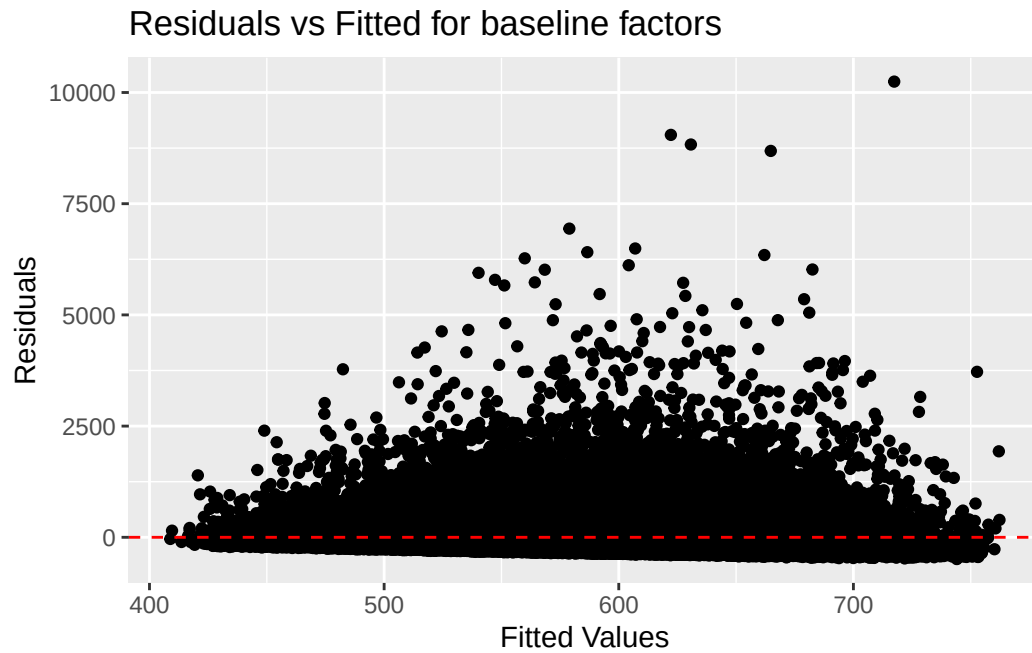
``stat_bin()`` using ``bins = 30``. Pick better value with ``binwidth``.



```
ggplot(data.frame(sample = residuals(mM)), aes(sample = sample)) +  
  stat_qq(color = "black") +  
  stat_qq_line(color = "red", linetype = "dashed") +  
  labs(title = "Q-Q Plot for modifiable lifestyle choices",  
        x = "Theoretical Quantiles", y = "Sample Quantiles")
```

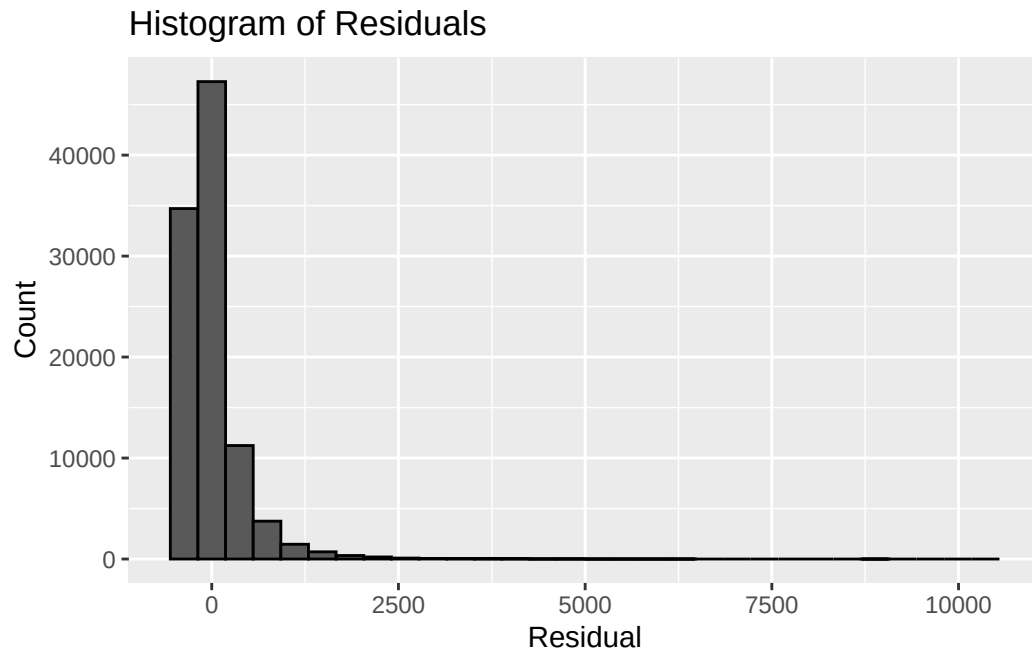


```
ggplot(data.frame(fitted = fitted(mB), residuals = residuals(mB)),  
  aes(x = fitted, y = residuals)) +  
  geom_point(color = "black") +  
  geom_hline(yintercept = 0, color = "red", linetype = "dashed") +  
  labs(title = "Residuals vs Fitted for baseline factors",  
    x = "Fitted Values", y = "Residuals")
```

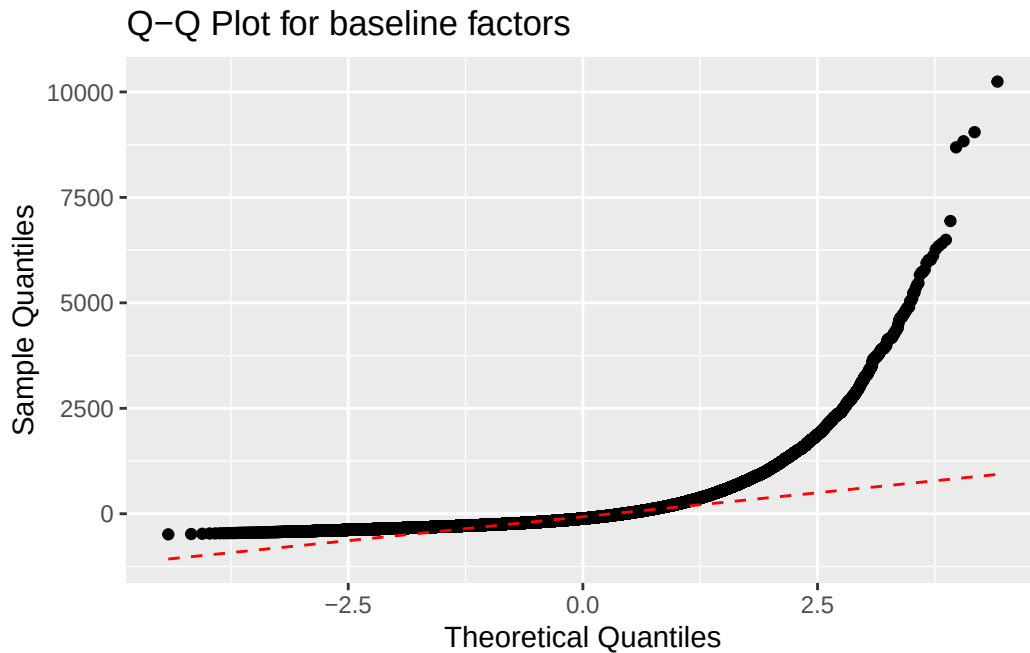


```
ggplot(data = mB, mapping = aes(x = mB$residuals)) +  
  geom_histogram(color = "black") +  
  labs(title = "Histogram of Residuals",  
        x = "Residual", y = "Count")
```

``stat_bin()`` using ``bins = 30``. Pick better value with ``binwidth``.



```
ggplot(data.frame(sample = residuals(mB)), aes(sample = sample)) +  
  stat_qq(color = "black") +  
  stat_qq_line(color = "red", linetype = "dashed") +  
  labs(title = "Q-Q Plot for baseline factors",  
        x = "Theoretical Quantiles", y = "Sample Quantiles")
```



For linear regression to be used, it must fulfill 4 requirements: 1) linearity in the parameters 2) constant variance of the errors 3) errors follow normal distribution 4) errors are independent.

Linearity: As observations are not distributed symmetrically about the y axis, it can be inferred that the assumption that linearity exists in the parameters is not fulfilled.

Constant variance of errors: as the residuals are not spaced out evenly (vertically), it can also be said that the variance of the errors is not constant.

Normal distribution of errors: The histogram of residuals is clearly right skewed, indicating that the errors are not normally distributed.

Independent Errors: As data is collected from a variety of sources, it is difficult to draw a conclusion here.

As linearity, constant variance of errors and normal distribution of errors is violated, we should use a generalized linear model (glm) rather than a linear model when working with this data.

```
mMgen <- glm(annual_premium ~ bmi + smoker + alcohol_freq, data=insurance, family = "Gamma")
summary(mMgen)
```

Call:

```
glm(formula = annual_premium ~ bmi + smoker + alcohol_freq, family = "Gamma",
    data = insurance)
```


Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.601e-03	2.629e-05	60.910	<2e-16 ***
bmi	-9.996e-06	7.181e-07	-13.920	<2e-16 ***
smokerFormer	3.164e-04	1.157e-05	27.357	<2e-16 ***
smokerNever	4.718e-04	9.347e-06	50.479	<2e-16 ***
alcohol_freqNone	1.537e-05	1.720e-05	0.894	0.371
alcohol_freqOccasional	1.723e-05	1.678e-05	1.027	0.305
alcohol_freqWeekly	2.586e-05	1.785e-05	1.449	0.147

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.4430359)

Null deviance: 27768 on 99999 degrees of freedom
Residual deviance: 26665 on 99993 degrees of freedom
AIC: 1401693

Number of Fisher Scoring iterations: 6

```
mBgen <- glm(annual_premium ~ age + sex + education + income + marital_status + household_size, data = insurance, family = "Gamma")
summary(mBgen)
```

Call:

```
glm(formula = annual_premium ~ age + sex + education + income + marital_status + household_size, family = "Gamma", data = insurance)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.185e-03	2.085e-05	104.769	<2e-16 ***
age	-9.320e-06	2.257e-07	-41.294	<2e-16 ***
sexMale	-7.249e-06	7.346e-06	-0.987	0.3238
sexOther	3.269e-06	2.630e-05	0.124	0.9011
educationDoctorate	2.136e-05	2.266e-05	0.943	0.3458
educationHS	-1.069e-05	1.005e-05	-1.064	0.2873
educationMasters	-2.049e-05	1.187e-05	-1.726	0.0843 .
educationNo HS	-3.653e-05	1.723e-05	-2.120	0.0340 *
educationSome College	-1.172e-05	1.002e-05	-1.170	0.2422
income	1.704e-10	7.882e-11	2.162	0.0306 *
marital_statusMarried	4.995e-06	1.510e-05	0.331	0.7408

```

marital_statusSingle 5.364e-06 1.503e-05 0.357 0.7212
marital_statusWidowed 9.903e-06 2.277e-05 0.435 0.6637
household_size -4.892e-06 3.813e-06 -1.283 0.1996
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for Gamma family taken to be 0.4518264)

```

Null deviance: 27768  on 99999  degrees of freedom
Residual deviance: 26999  on 99986  degrees of freedom
AIC: 1403005

```

Number of Fisher Scoring iterations: 6

```

# 1.DATA ANALYSIS
# View data structure
glimpse(insurance)

```

```

Rows: 100,000
Columns: 54
$ person_id      <int> 75722, 80185, 19865, 76700, 92992, 76435, ~
$ age            <int> 52, 79, 68, 15, 53, 63, 36, 21, 53, 28, 34~
$ sex            <chr> "Female", "Female", "Male", "Male", "Male"~
$ region         <chr> "North", "North", "North", "North", "Centr~
$ urban_rural    <chr> "Suburban", "Urban", "Rural", "Suburban", ~
$ income         <dbl> 22700, 12800, 40700, 15600, 89600, 305000,~
$ education      <chr> "Doctorate", "No HS", "HS", "Some College"~
$ marital_status <chr> "Married", "Married", "Married", "Married"~
$ employment_status <chr> "Retired", "Employed", "Retired", "Self-em~
$ household_size <int> 3, 3, 5, 5, 2, 3, 1, 3, 1, 4, 2, 2, 3, 4, ~
$ dependents     <int> 1, 1, 3, 3, 0, 2, 0, 2, 0, 2, 1, 0, 1, 2, ~
$ bmi            <dbl> 27.4, 26.6, 31.5, 31.6, 30.5, 20.4, 21.6, ~
$ smoker         <chr> "Never", "Never", "Never", "Never", "Never~
$ alcohol_freq   <chr> "None", "Weekly", "None", "None", "Daily",~
$ visits_last_year <int> 2, 2, 1, 0, 3, 1, 0, 3, 0, 0, 8, 2, 2, 1, ~
$ hospitalizations_last_3yrs <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ days_hospitalized_last_3yrs <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ medication_count <int> 4, 3, 4, 1, 2, 2, 0, 1, 0, 0, 3, 3, 0, 2, ~
$ systolic_bp    <dbl> 121, 131, 160, 104, 136, 137, 114, 82, 138~
$ diastolic_bp   <dbl> 76, 79, 84, 68, 83, 96, 76, 64, 84, 62, 65~
$ ldl            <dbl> 123.8, 97.3, 129.5, 160.3, 171.0, 69.2, 15~
$ hba1c          <dbl> 5.28, 4.82, 5.51, 8.50, 5.20, 5.70, 5.54, ~

```

```

$ plan_type          <chr> "PPO", "POS", "HMO", "HMO", "POS", "HMO", ~
$ network_tier       <chr> "Bronze", "Gold", "Platinum", "Silver", "P~
$ deductible         <int> 1000, 1000, 500, 500, 500, 500, 500, 2000,~
$ copay              <int> 20, 10, 20, 20, 10, 20, 10, 20, 10, 10, 20~
$ policy_term_years  <int> 4, 1, 10, 5, 7, 5, 3, 6, 4, 4, 10, 9, 5, 2~
$ policy_changes_last_2yrs <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ provider_quality   <dbl> 3.73, 3.10, 3.90, 3.89, 3.90, 4.66, 4.30, ~
$ risk_score         <dbl> 0.5714, 1.0000, 1.0000, 0.2857, 0.8681, 0.~
$ annual_medical_cost <dbl> 6938.06, 1632.61, 7661.01, 5130.27, 1700.7~
$ annual_premium     <dbl> 876.05, 445.10, 1538.02, 820.63, 500.93, 5~
$ monthly_premium    <dbl> 73.00, 37.09, 128.17, 68.39, 41.74, 44.06,~
$ claims_count       <int> 1, 4, 0, 0, 1, 1, 0, 2, 0, 0, 11, 2, 1, 3,~
$ avg_claim_amount   <dbl> 4672.59, 297.27, 0.00, 0.00, 1002.24, 1220~
$ total_claims_paid  <dbl> 4672.59, 1189.08, 0.00, 0.00, 1002.24, 122~
$ chronic_count      <int> 1, 2, 3, 1, 2, 1, 0, 1, 1, 0, 2, 1, 0, 0, ~
$ hypertension       <int> 0, 0, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, ~
$ diabetes           <int> 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
$ asthma             <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ copd               <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ cardiovascular_disease <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, ~
$ cancer_history     <int> 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ kidney_disease     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ liver_disease      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ arthritis          <int> 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, ~
$ mental_health      <int> 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, ~
$ proc_imaging_count <int> 1, 0, 1, 1, 2, 0, 1, 2, 0, 0, 0, 1, 0, 1, ~
$ proc_surgery_count <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~
$ proc_physio_count  <int> 2, 1, 2, 0, 1, 0, 1, 0, 1, 2, 1, 1, 1, 0, ~
$ proc_consult_count <int> 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, ~
$ proc_lab_count     <int> 1, 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 1, 2, 0, ~
$ is_high_risk       <int> 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, ~
$ had_major_procedure <int> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ~

```

```
summary(insurance)
```

person_id	age	sex	region
Min. : 1	Min. : 0.00	Length:100000	Length:100000
1st Qu.: 25001	1st Qu.: 37.00	Class :character	Class :character
Median : 50000	Median : 48.00	Mode :character	Mode :character
Mean : 50000	Mean : 47.52		
3rd Qu.: 75000	3rd Qu.: 58.00		
Max. :100000	Max. :100.00		

urban_rural	income	education	marital_status
Length:100000	Min. : 1100	Length:100000	Length:100000
Class :character	1st Qu.: 21100	Class :character	Class :character
Mode :character	Median : 36200	Mode :character	Mode :character
	Mean : 49874		
	3rd Qu.: 62200		
	Max. :1061800		
employment_status	household_size	dependents	bmi
Length:100000	Min. :1.000	Min. :0.0000	Min. :12.00
Class :character	1st Qu.:2.000	1st Qu.:0.0000	1st Qu.:23.60
Mode :character	Median :2.000	Median :1.0000	Median :27.00
	Mean :2.431	Mean :0.8984	Mean :26.99
	3rd Qu.:3.000	3rd Qu.:1.0000	3rd Qu.:30.40
	Max. :9.000	Max. :7.0000	Max. :50.40
smoker	alcohol_freq	visits_last_year	
Length:100000	Length:100000	Min. : 0.000	
Class :character	Class :character	1st Qu.: 1.000	
Mode :character	Mode :character	Median : 2.000	
		Mean : 1.928	
		3rd Qu.: 3.000	
		Max. :25.000	
hospitalizations_last_3yrs	days_hospitalized_last_3yrs	medication_count	
Min. :0.00000	Min. : 0.0000	Min. : 0.000	
1st Qu.:0.00000	1st Qu.: 0.0000	1st Qu.: 0.000	
Median :0.00000	Median : 0.0000	Median : 1.000	
Mean :0.09364	Mean : 0.3734	Mean : 1.236	
3rd Qu.:0.00000	3rd Qu.: 0.0000	3rd Qu.: 2.000	
Max. :3.00000	Max. :21.0000	Max. :11.000	
systolic_bp	diastolic_bp	ldl	hba1c
Min. : 61.0	Min. : 40.0	Min. : 30.0	Min. : 3.540
1st Qu.:107.0	1st Qu.: 67.0	1st Qu.: 99.4	1st Qu.: 5.160
Median :117.0	Median : 73.0	Median :120.0	Median : 5.440
Mean :117.8	Mean : 73.6	Mean :120.0	Mean : 5.606
3rd Qu.:128.0	3rd Qu.: 79.0	3rd Qu.:140.3	3rd Qu.: 5.760
Max. :183.0	Max. :114.0	Max. :248.3	Max. :11.940
plan_type	network_tier	deductible	copay
Length:100000	Length:100000	Min. : 500	Min. :10.00
Class :character	Class :character	1st Qu.: 500	1st Qu.:10.00
Mode :character	Mode :character	Median :1000	Median :20.00
		Mean :1227	Mean :19.52
		3rd Qu.:2000	3rd Qu.:30.00
		Max. :5000	Max. :50.00
policy_term_years	policy_changes_last_2yrs	provider_quality	risk_score

Min. : 1.000	Min. :0.00000	Min. :1.500	Min. :0.0000
1st Qu.: 3.000	1st Qu.:0.00000	1st Qu.:3.200	1st Qu.:0.3297
Median : 6.000	Median :0.00000	Median :3.600	Median :0.5055
Mean : 5.518	Mean :0.05069	Mean :3.599	Mean :0.5198
3rd Qu.: 8.000	3rd Qu.:0.00000	3rd Qu.:4.010	3rd Qu.:0.7033
Max. :10.000	Max. :2.00000	Max. :5.000	Max. :1.0000
annual_medical_cost	annual_premium	monthly_premium	claims_count
Min. : 55.55	Min. : 211.7	Min. : 17.64	Min. : 0.000
1st Qu.: 1175.12	1st Qu.: 352.1	1st Qu.: 29.34	1st Qu.: 0.000
Median : 2082.57	Median : 463.6	Median : 38.63	Median : 1.000
Mean : 3009.45	Mean : 582.3	Mean : 48.53	Mean : 1.622
3rd Qu.: 3707.96	3rd Qu.: 666.7	3rd Qu.: 55.56	3rd Qu.: 2.000
Max. :65724.90	Max. :10962.5	Max. :913.55	Max. :23.000
avg_claim_amount	total_claims_paid	chronic_count	hypertension
Min. : 0.0	Min. : 0.0	Min. :0.0000	Min. :0.0000
1st Qu.: 0.0	1st Qu.: 0.0	1st Qu.:0.0000	1st Qu.:0.0000
Median : 318.0	Median : 642.5	Median :1.0000	Median :0.0000
Mean : 656.5	Mean : 1377.9	Mean :0.7247	Mean :0.2034
3rd Qu.: 872.2	3rd Qu.: 1795.5	3rd Qu.:1.0000	3rd Qu.:0.0000
Max. :30010.5	Max. :72517.9	Max. :6.0000	Max. :1.0000
diabetes	asthma	copd	cardiovascular_disease
Min. :0.00000	Min. :0.00000	Min. :0.00000	Min. :0.00000
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.00000
Median :0.00000	Median :0.00000	Median :0.00000	Median :0.00000
Mean :0.08593	Mean :0.05887	Mean :0.03595	Mean :0.05117
3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.00000
Max. :1.00000	Max. :1.00000	Max. :1.00000	Max. :1.00000
cancer_history	kidney_disease	liver_disease	arthritis
Min. :0.00000	Min. :0.00000	Min. :0.00000	Min. :0.0000
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.0000
Median :0.00000	Median :0.00000	Median :0.00000	Median :0.0000
Mean :0.02151	Mean :0.01462	Mean :0.01477	Mean :0.1083
3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:0.0000
Max. :1.00000	Max. :1.00000	Max. :1.00000	Max. :1.0000
mental_health	proc_imaging_count	proc_surgery_count	proc_physio_count
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.1301	Mean :0.5085	Mean :0.1587	Mean :0.5084
3rd Qu.:0.0000	3rd Qu.:1.0000	3rd Qu.:0.0000	3rd Qu.:1.0000
Max. :1.0000	Max. :7.0000	Max. :6.0000	Max. :7.0000
proc_consult_count	proc_lab_count	is_high_risk	had_major_procedure
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000

1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.5093	Mean :0.5091	Mean :0.3678	Mean :0.1697
3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:0.0000
Max. :7.0000	Max. :7.0000	Max. :1.0000	Max. :1.0000

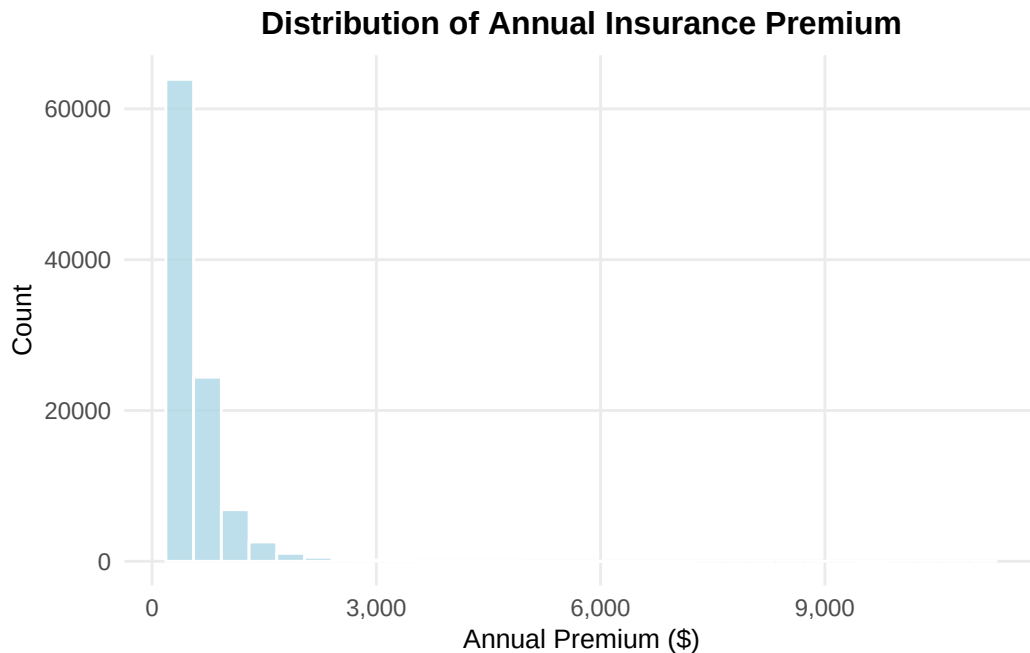
```
# Check for missing values
colSums(is.na(insurance))
```

person_id	age
0	0
sex	region
0	0
urban_rural	income
0	0
education	marital_status
0	0
employment_status	household_size
0	0
dependents	bmi
0	0
smoker	alcohol_freq
0	0
visits_last_year	hospitalizations_last_3yrs
0	0
days_hospitalized_last_3yrs	medication_count
0	0
systolic_bp	diastolic_bp
0	0
ldl	hba1c
0	0
plan_type	network_tier
0	0
deductible	copay
0	0
policy_term_years	policy_changes_last_2yrs
0	0
provider_quality	risk_score
0	0
annual_medical_cost	annual_premium
0	0
monthly_premium	claims_count

0	0
avg_claim_amount	total_claims_paid
0	0
chronic_count	hypertension
0	0
diabetes	asthma
0	0
copd	cardiovascular_disease
0	0
cancer_history	kidney_disease
0	0
liver_disease	arthritis
0	0
mental_health	proc_imaging_count
0	0
proc_surgery_count	proc_physio_count
0	0
proc_consult_count	proc_lab_count
0	0
is_high_risk	had_major_procedure
0	0

```
# univariate visualizations

# Distribution of (annual_premium) using a histogram
ggplot(insurance, aes(x = annual_premium)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white", alpha = 0.8) +
  labs(title = "Distribution of Annual Insurance Premium",
       x = "Annual Premium ($)",
       y = "Count") +
  scale_x_continuous(labels = scales::comma) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
        panel.grid.minor = element_blank())
```



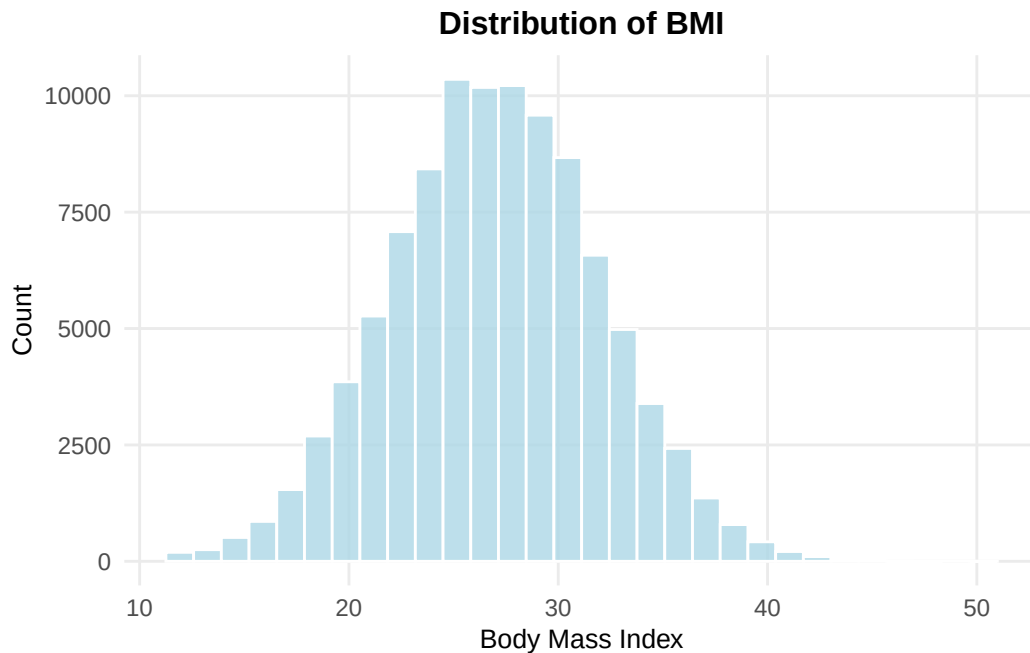
```
# Summary statistics for premium
summary(insurance$annual_premium)
```

```

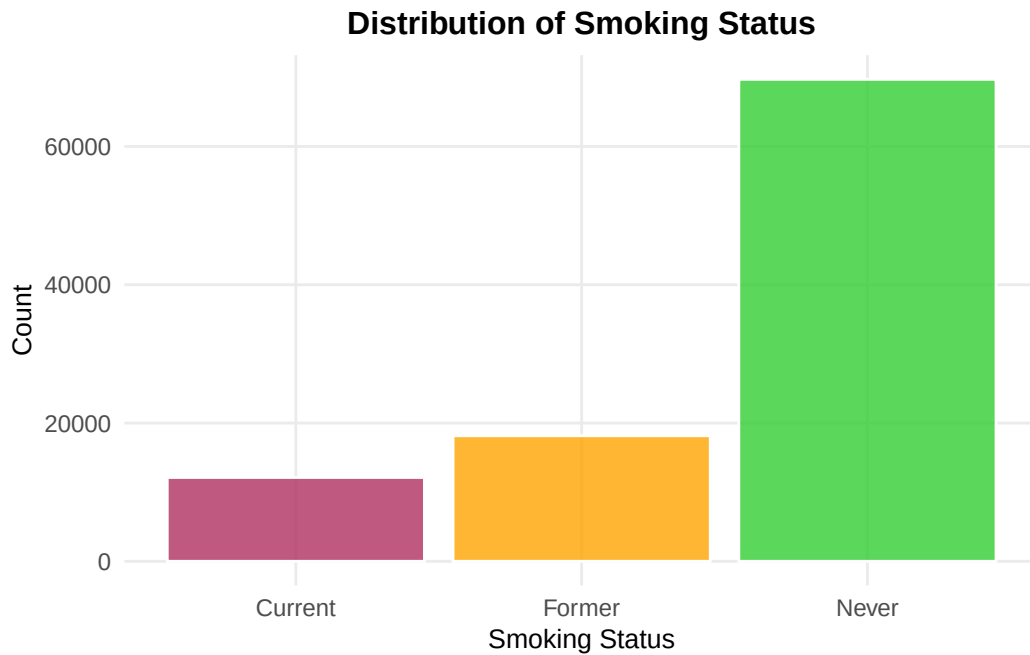
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 211.7   352.1   463.6   582.3   666.7 10962.5

```

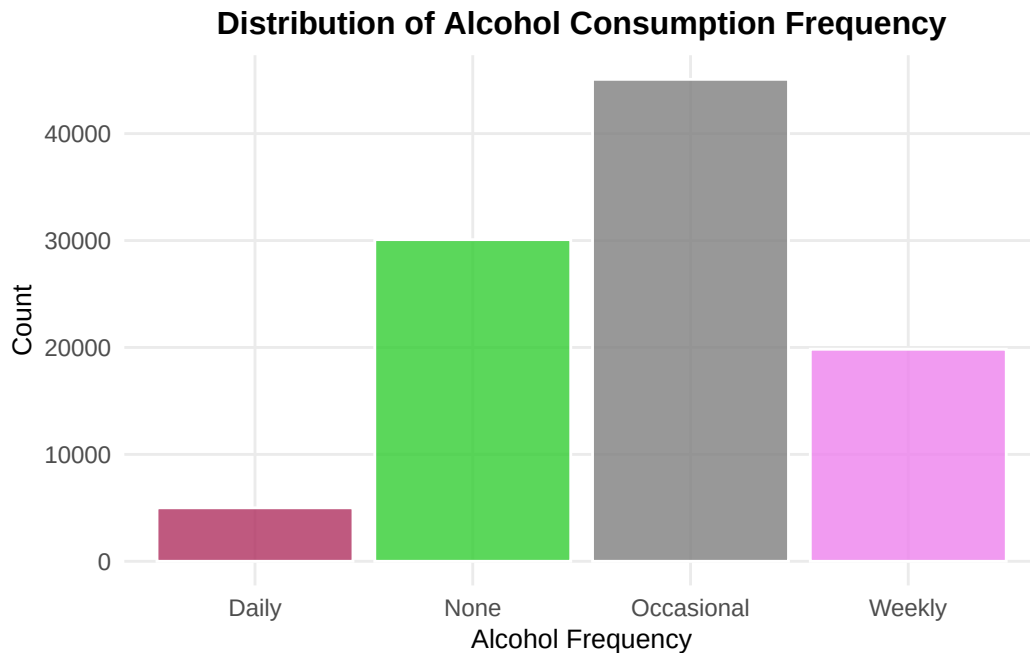
```
# visualizations for distributions of modifiable lifestyle factors
# BMI visual
ggplot(insurance, aes(x = bmi)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white", alpha = 0.8) +
  labs(title = "Distribution of BMI",
       x = "Body Mass Index",
       y = "Count") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
        panel.grid.minor = element_blank())
```

```
# Smoking Status
ggplot(insurance, aes(x = smoker, fill = smoker)) +
  geom_bar(alpha = 0.8, color = "white") +
  scale_fill_manual(values = c("Current" = "maroon",
                                "Former" = "orange",
                                "Never" = "limegreen")) +
  labs(title = "Distribution of Smoking Status",
       x = "Smoking Status",
       y = "Count") +
  theme_minimal() +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
        panel.grid.minor = element_blank())
```



```
# Alcohol Frequency
ggplot(insurance, aes(x = alcohol_freq, fill = alcohol_freq)) +
  geom_bar(alpha = 0.8, color = "white") +
  scale_fill_manual(values = c("Daily" = "maroon",
                                "None" = "limegreen",
                                "Occasionally" = "orange",
                                "Weekly" = "violet")) +
  labs(title = "Distribution of Alcohol Consumption Frequency",
       x = "Alcohol Frequency",
       y = "Count") +
  theme_minimal() +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
        axis.title = element_text(size = 10),
        panel.grid.minor = element_blank())
```



```
# BIVARIATE STATISTICAL TESTS
```

```
# BMI correlation with premium
```

```
cat("\n Correlation Test: BMI vs Annual Premium \n")
```

Correlation Test: BMI vs Annual Premium

```
cor_test_bmi <- cor.test(insurance$bmi, insurance$annual_premium)
print(cor_test_bmi)
```

Pearson's product-moment correlation

```
data:  insurance$bmi and insurance$annual_premium
t = 13.19, df = 99998, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.03548612 0.04786052
sample estimates:
      cor
0.04167491
```

```
# ANOVA for Smoking Status
cat("\n ANOVA: Smoking Status vs Annual Premium \n")
```

ANOVA: Smoking Status vs Annual Premium

```
anova_smoking <- aov(annual_premium ~ smoker, data = insurance)
print(summary(anova_smoking))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
smoker	2	3.884e+08	194204593	1247	<2e-16 ***
Residuals	99997	1.558e+10	155786		

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# ANOVA for Alcohol Frequency
cat("\n ANOVA: Alcohol Frequency vs Annual Premium \n")
```

ANOVA: Alcohol Frequency vs Annual Premium

```
anova_alcohol <- aov(annual_premium ~ alcohol_freq, data = insurance)
print(summary(anova_alcohol))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
alcohol_freq	3	2.860e+05	95318	0.597	0.617
Residuals	99996	1.597e+10	159669		

Initial bivariate analyses revealed differential relationships between modifiable lifestyle factors and annual insurance premiums. BMI demonstrated a statistically significant but weak positive correlation with premium costs ($r = 0.042$, 95% CI [0.035, 0.048], $p < 0.001$), suggesting that BMI alone explains minimal variance in insurance costs. In contrast, smoking status showed a substantial and highly significant association with premiums ($F(2, 99997) = 1247$, $p < 0.001$), indicating meaningful differences in insurance costs across smoking categories. Notably, alcohol consumption frequency was not significantly associated with premium costs ($F(3, 99996) = 0.597$, $p = 0.617$), suggesting that alcohol use patterns do not independently predict insurance premiums in this sample. These preliminary findings suggest that while smoking status is a strong univariate predictor of insurance costs, the relationship between BMI and premiums may be more complex and potentially moderated by other factors, warranting multivariate analysis to account for confounding variables and interactions.

```

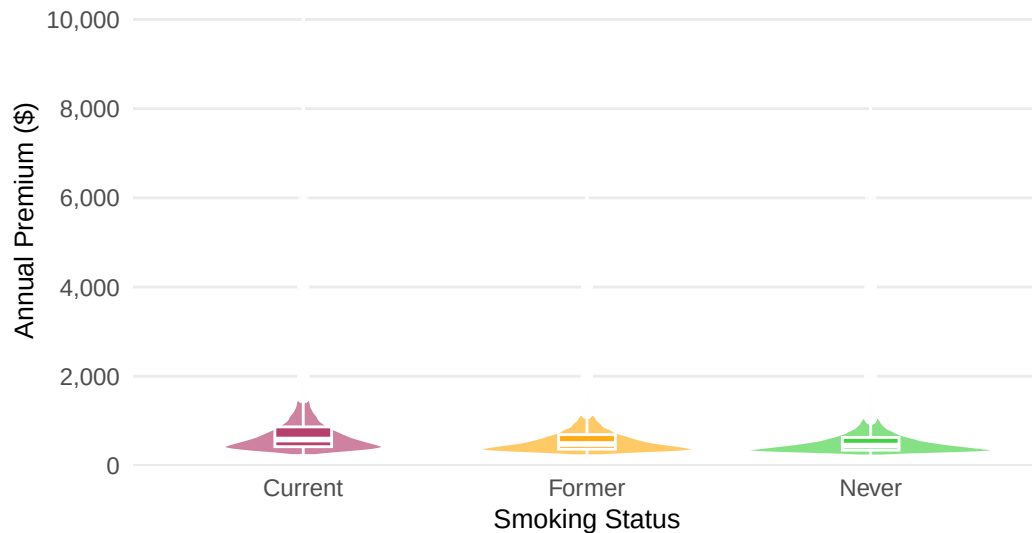
ggplot(insurance, aes(x = smoker, y = annual_premium, fill = smoker)) +
  geom_violin(alpha = 0.6, color = "white") +
  geom_boxplot(width = 0.2, alpha = 0.8, color = "white", outlier.alpha = 0.5) +
  scale_fill_manual(values = c("Current" = "maroon",
                                "Former" = "orange",
                                "Never" = "limegreen")) +

  scale_y_continuous(
    labels = scales::comma,
    breaks = seq(0, 10000, 2000),
    expand = c(0, 0.02),
    limits = c(0, 11000)
  ) +
  labs(title = "Insurance Premium by Smoking Status",
       subtitle = "Showing the primary effect of smoking on premiums",
       x = "Smoking Status",
       y = "Annual Premium ($)") +
  theme_minimal() +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
    plot.subtitle = element_text(size = 10, hjust = 0.5),
    axis.title = element_text(size = 10),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank()
  )

```

Insurance Premium by Smoking Status

Showing the primary effect of smoking on premiums



As earlier plot may be difficult to see, we create a second plot where limits from 0 to 2000

```
ggplot(insurance, aes(x = smoker, y = annual_premium, fill = smoker)) +
  geom_violin(alpha = 0.6, color = "white") +
  geom_boxplot(width = 0.2, alpha = 0.8, color = "white", outlier.alpha = 0.5) +
  scale_fill_manual(values = c("Current" = "maroon",
                                "Former" = "orange",
                                "Never" = "limegreen")) +

  scale_y_continuous(
    labels = scales::comma,
    breaks = seq(0, 10000, 2000),
    expand = c(0, 0.02),
    limits = c(0, 2000)
  ) +
  labs(title = "Insurance Premium by Smoking Status (0 < Annual Premium < 2000)",
       subtitle = "Showing the primary effect of smoking on premiums",
       x = "Smoking Status",
       y = "Annual Premium ($)") +
  theme_minimal() +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
    plot.subtitle = element_text(size = 10, hjust = 0.5),
```

```
axis.title = element_text(size = 10),
panel.grid.minor = element_blank(),
panel.grid.major.x = element_blank()
)
```

Warning: Removed 1294 rows containing non-finite outside the scale range (`stat\_ydensity()`).

Warning: Removed 1294 rows containing non-finite outside the scale range (`stat\_boxplot()`).

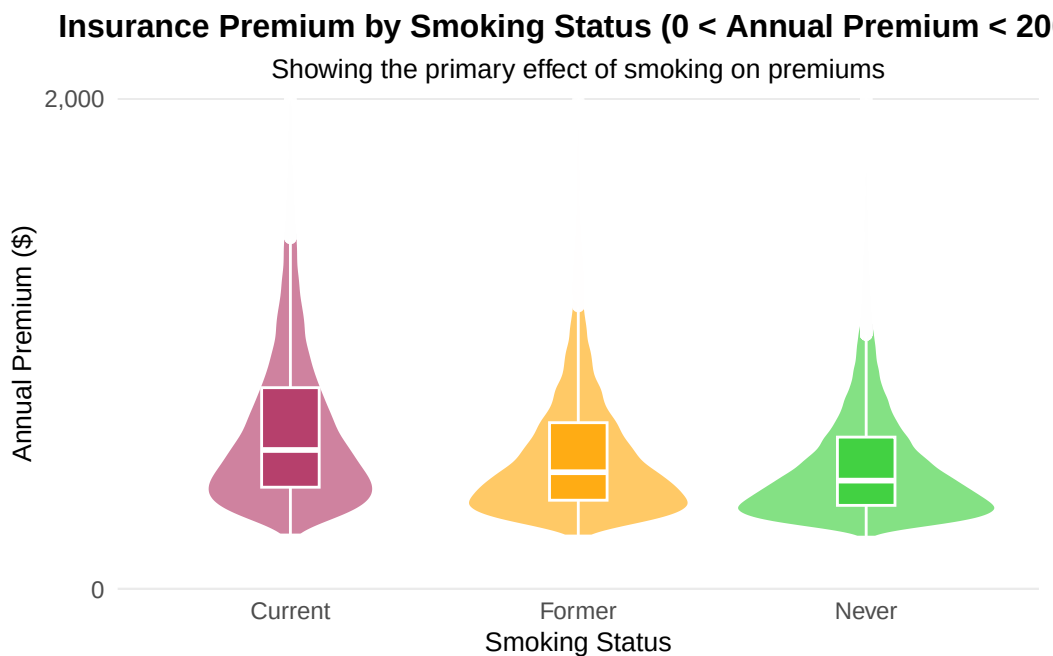


Figure [X] displays the distribution of annual insurance premiums across smoking status categories. The violin plots reveal substantial differences in premium distributions among the three groups. Current smokers demonstrate notably elevated premiums with a median around 1,200 USD and considerable variability extending to over 1,500 USD for the upper quartile. Former smokers show an intermediate distribution, with lower median premiums than current smokers but higher than never-smokers. Never-smokers exhibit the lowest and most concentrated premium distribution, with the majority of observations clustering below 1,000 USD. The pronounced vertical separation between groups—particularly between current smokers and never-smokers—provides preliminary visual evidence that smoking status, a modifiable

lifestyle factor, is strongly associated with insurance costs. The overlapping distributions suggest some shared variance with other factors, which will be explored through formal variance partitioning analysis.

```
# regression model stuff

#Modifiable lifestyle factors only
m1 <- glm(annual_premium ~ bmi + smoker + alcohol_freq, data = insurance)

#Baseline demographic/socioeconomic factors only
m2 <- glm(annual_premium ~ age + sex + education + income +
          marital_status + household_size, data = insurance)

#Full model (both modifiable and baseline factors)
m3 <- glm(annual_premium ~ bmi + smoker + alcohol_freq + age + sex +
          education + income + marital_status + household_size, data = insurance)

#Full model with interaction (BMI * smoker)
m4 <- glm(annual_premium ~ bmi * smoker + alcohol_freq + age + sex +
          education + income + marital_status + household_size, data = insurance)

#Display model summaries
cat("\n MODEL 1: Modifiable Factors Only \n")
```

MODEL 1: Modifiable Factors Only

```
summary(m1)
```

Call:

```
glm(formula = annual_premium ~ bmi + smoker + alcohol_freq, data = insurance)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	655.3464	9.3629	69.994	<2e-16 ***
bmi	3.4341	0.2497	13.754	<2e-16 ***
smokerFormer	-141.0053	4.6245	-30.491	<2e-16 ***
smokerNever	-192.4150	3.8799	-49.593	<2e-16 ***
alcohol_freqNone	-5.2296	6.0193	-0.869	0.385
alcohol_freqOccasional	-5.8825	5.8747	-1.001	0.317


```

alcohol_freqWeekly      -8.8204      6.2373  -1.414    0.157
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 155494.2)

Null deviance: 1.5967e+10  on 99999  degrees of freedom
Residual deviance: 1.5548e+10  on 99993  degrees of freedom
AIC: 1479233

Number of Fisher Scoring iterations: 2

```

```
cat("\n MODEL 2: Baseline Factors Only \n")
```

MODEL 2: Baseline Factors Only

```
summary(m2)
```

```

Call:
glm(formula = annual_premium ~ age + sex + education + income +
    marital_status + household_size, data = insurance)

```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.268e+02	7.089e+00	60.210	<2e-16	***
age	3.186e+00	7.839e-02	40.638	<2e-16	***
sexMale	2.538e+00	2.532e+00	1.002	0.3162	
sexOther	-1.051e+00	9.012e+00	-0.117	0.9072	
educationDoctorate	-7.176e+00	7.663e+00	-0.936	0.3490	
educationHS	3.658e+00	3.455e+00	1.059	0.2897	
educationMasters	7.065e+00	4.104e+00	1.722	0.0851	.
educationNo HS	1.285e+01	6.024e+00	2.132	0.0330	*
educationSome College	3.972e+00	3.445e+00	1.153	0.2489	
income	-5.740e-05	2.678e-05	-2.143	0.0321	*
marital_statusMarried	-1.665e+00	5.213e+00	-0.319	0.7494	
marital_statusSingle	-1.915e+00	5.185e+00	-0.369	0.7119	
marital_statusWidowed	-3.340e+00	7.829e+00	-0.427	0.6697	
household_size	1.690e+00	1.318e+00	1.282	0.1998	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 157067.4)

Null deviance: 1.5967e+10 on 99999 degrees of freedom
Residual deviance: 1.5705e+10 on 99986 degrees of freedom
AIC: 1480247

Number of Fisher Scoring iterations: 2

```
cat("\n MODEL 3: Full Model \n")
```

MODEL 3: Full Model

```
summary(m3)
```

Call:

```
glm(formula = annual_premium ~ bmi + smoker + alcohol_freq +  
    age + sex + education + income + marital_status + household_size,  
    data = insurance)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.996e+02	1.155e+01	43.239	<2e-16 ***
bmi	3.412e+00	2.475e-01	13.784	<2e-16 ***
smokerFormer	-1.412e+02	4.585e+00	-30.785	<2e-16 ***
smokerNever	-1.936e+02	3.847e+00	-50.318	<2e-16 ***
alcohol_freqNone	-4.573e+00	5.968e+00	-0.766	0.4436
alcohol_freqOccasional	-5.761e+00	5.825e+00	-0.989	0.3226
alcohol_freqWeekly	-9.452e+00	6.184e+00	-1.528	0.1264
age	3.214e+00	7.734e-02	41.562	<2e-16 ***
sexMale	3.333e+00	2.498e+00	1.334	0.1821
sexOther	-9.985e-01	8.891e+00	-0.112	0.9106
educationDoctorate	-7.615e+00	7.559e+00	-1.007	0.3138
educationHS	3.581e+00	3.408e+00	1.051	0.2934
educationMasters	7.980e+00	4.048e+00	1.971	0.0487 *
educationNo HS	1.322e+01	5.943e+00	2.224	0.0261 *
educationSome College	4.348e+00	3.398e+00	1.280	0.2007
income	-5.671e-05	2.642e-05	-2.146	0.0318 *

```
marital_statusMarried -1.934e+00  5.143e+00 -0.376  0.7069
marital_statusSingle -1.619e+00  5.115e+00 -0.317  0.7516
marital_statusWidowed -5.039e+00  7.723e+00 -0.652  0.5141
household_size        1.556e+00  1.301e+00  1.197  0.2315
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

(Dispersion parameter for gaussian family taken to be 152845)

```
Null deviance: 1.5967e+10  on 99999  degrees of freedom
Residual deviance: 1.5281e+10  on 99980  degrees of freedom
AIC: 1477528
```

Number of Fisher Scoring iterations: 2

```
cat("\n MODEL 4: Full Model + Interaction \n")
```

MODEL 4: Full Model + Interaction

```
summary(m4)
```

Call:

```
glm(formula = annual_premium ~ bmi * smoker + alcohol_freq +
     age + sex + education + income + marital_status + household_size,
     data = insurance)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.864e+02	2.123e+01	22.913	< 2e-16 ***
bmi	3.904e+00	7.100e-01	5.499	3.82e-08 ***
smokerFormer	-1.214e+02	2.513e+01	-4.831	1.36e-06 ***
smokerNever	-1.797e+02	2.103e+01	-8.547	< 2e-16 ***
alcohol_freqNone	-4.596e+00	5.968e+00	-0.770	0.4413
alcohol_freqOccasional	-5.779e+00	5.825e+00	-0.992	0.3212
alcohol_freqWeekly	-9.475e+00	6.184e+00	-1.532	0.1255
age	3.214e+00	7.734e-02	41.558	< 2e-16 ***
sexMale	3.321e+00	2.498e+00	1.329	0.1838
sexOther	-1.032e+00	8.891e+00	-0.116	0.9076
educationDoctorate	-7.647e+00	7.559e+00	-1.012	0.3117

educationHS	3.587e+00	3.408e+00	1.052	0.2927
educationMasters	7.979e+00	4.048e+00	1.971	0.0487 *
educationNo HS	1.324e+01	5.943e+00	2.227	0.0259 *
educationSome College	4.343e+00	3.398e+00	1.278	0.2012
income	-5.674e-05	2.642e-05	-2.148	0.0317 *
marital_statusMarried	-1.934e+00	5.143e+00	-0.376	0.7069
marital_statusSingle	-1.609e+00	5.115e+00	-0.314	0.7531
marital_statusWidowed	-5.056e+00	7.723e+00	-0.655	0.5127
household_size	1.553e+00	1.301e+00	1.194	0.2325
bmi:smokerFormer	-7.345e-01	9.171e-01	-0.801	0.4232
bmi:smokerNever	-5.149e-01	7.694e-01	-0.669	0.5033

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 152847)

Null deviance: 1.5967e+10 on 99999 degrees of freedom
 Residual deviance: 1.5281e+10 on 99978 degrees of freedom
 AIC: 1477531

Number of Fisher Scoring iterations: 2

We fit four linear regression models to examine how modifiable lifestyle factors and baseline demographic characteristics relate to insurance premiums. Model 1 included only modifiable factors (BMI, smoking status, alcohol frequency), Model 2 included only baseline factors (age, sex, education, income, marital status, household size), Model 3 combined both sets of predictors, and Model 4 added a BMI by smoking interaction term.

Model 1 explained 2.6% of premium variance and was statistically significant ($F = 448.3$, $p < 0.001$). Smoking status was the strongest predictor. Compared to current smokers, former smokers paid \$141 less annually and never smokers paid \$192 less (both $p < 0.001$), representing roughly 20 to 26% lower premiums. BMI showed a positive association, with each unit increase linked to a \$3.43 premium increase ($p < 0.001$). Alcohol consumption was not significantly related to premiums at any level (all $p > 0.15$).

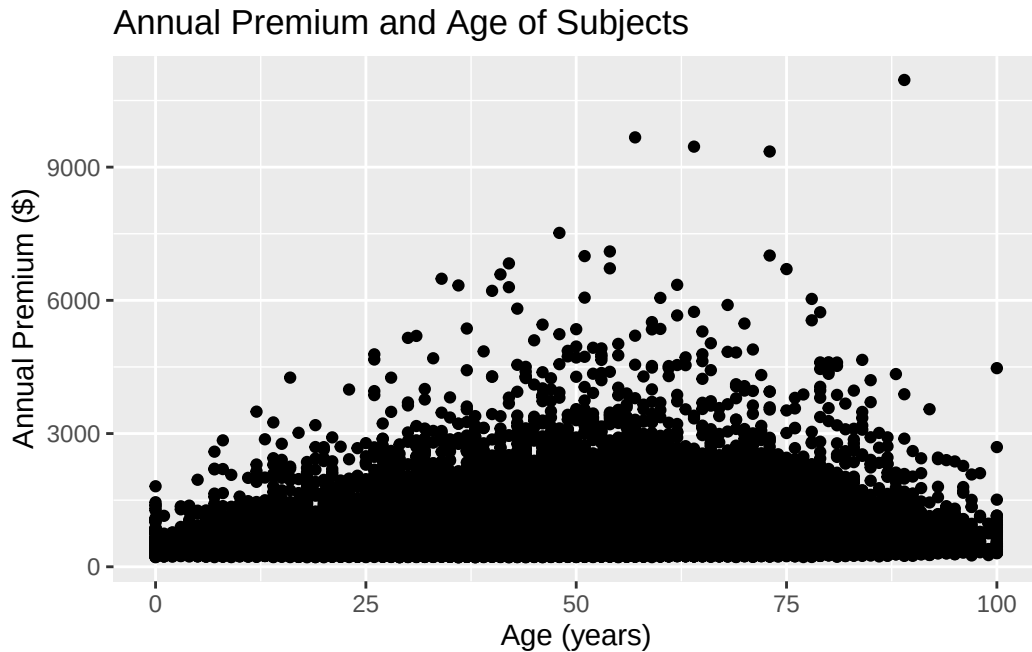
Model 2 explained only 1.6% of premium variance. Age was the strongest baseline predictor, with each year associated with a \$3.19 premium increase ($p < 0.001$). Income showed a small negative effect ($p = 0.032$), while sex, marital status, and household size were not significant.

Model 3 combined all predictors and explained 4.3% of variance, nearly the sum of Models 1 and 2. Key predictors maintained stable effect sizes. Smoking coefficients were nearly identical to Model 1 (former: \$141 less, never: \$194 less, both $p < 0.001$), as were BMI and age effects. This stability suggests minimal confounding between modifiable and baseline factors.

Model 4 tested for BMI by smoking interactions but explained virtually the same variance as Model 3 (4.29% vs 4.27%). The interaction terms were not significant ($p = 0.42$ and $p = 0.50$), indicating the relationship between BMI and premiums does not differ across smoking groups. We therefore selected Model 3 as our final model.

The low R-squared values across all models (1.6% to 4.3%) indicate that most premium variance is unexplained by measured predictors. This likely reflects unmeasured factors such as coverage type, deductibles, policy terms, or company-specific pricing. However, the significant effects of modifiable factors, particularly smoking status, demonstrate that lifestyle choices contribute meaningfully to premium determination.

```
ggplot(insurance, aes(x = age, y = annual_premium)) +  
  geom_point() +  
  labs(title = "Annual Premium and Age of Subjects",  
        x = "Age (years)",  
        y = "Annual Premium ($)")
```



Linear association between age and annual premium can be observed. This plot does not control for other variables.

```
# Sample characteristics  
cat("\n SAMPLE CHARACTERISTICS \n")
```

SAMPLE CHARACTERISTICS

```
cat(sprintf("Total observations: %d\n", nrow(insurance)))
```

Total observations: 100000

```
cat(sprintf("Complete cases: %d\n", sum(complete.cases(insurance))))
```

Complete cases: 100000

```
# Formal model comparison tests
cat("\n FORMAL MODEL COMPARISONS \n")
```

FORMAL MODEL COMPARISONS

```
# Test 1: Does adding modifiable factors improve baseline model?
cat("\nTest 1: Baseline Model (M2) vs Full Model (M3)\n")
```

Test 1: Baseline Model (M2) vs Full Model (M3)

```
cat("Question: Do modifiable factors add significant explanatory power?\n\n")
```

Question: Do modifiable factors add significant explanatory power?

```
anova_m2_m3 <- anova(m2, m3)
print(anova_m2_m3)
```

Analysis of Deviance Table

Model 1: annual_premium ~ age + sex + education + income + marital_status +
household_size

Model 2: annual_premium ~ bmi + smoker + alcohol_freq + age + sex + education +
income + marital_status + household_size

	Resid. Df	Resid. Dev	Df	Deviance	F	Pr(>F)
1	99986	1.5705e+10				
2	99980	1.5281e+10	6	423104569	461.37	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Test 2: Does adding baseline factors improve modifiable model?
cat("\nTest 2: Modifiable Model (M1) vs Full Model (M3)\n")
```

Test 2: Modifiable Model (M1) vs Full Model (M3)

```
cat("Question: Do baseline factors add significant explanatory power?\n\n")
```

Question: Do baseline factors add significant explanatory power?

```
anova_m1_m3 <- anova(m1, m3)
print(anova_m1_m3)
```

Analysis of Deviance Table

Model 1: annual_premium ~ bmi + smoker + alcohol_freq

Model 2: annual_premium ~ bmi + smoker + alcohol_freq + age + sex + education +
income + marital_status + household_size

	Resid. Df	Resid. Dev	Df	Deviance	F	Pr(>F)
1	99993	1.5548e+10				
2	99980	1.5281e+10	13	266897262	134.32	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Test 3: Is the interaction term significant?
cat("\nTest 3: Full Model (M3) vs Full + Interaction (M4)\n")
```

Test 3: Full Model (M3) vs Full + Interaction (M4)

```
cat("Question: Is the BMI × Smoking interaction statistically significant?\n\n")
```

Question: Is the BMI × Smoking interaction statistically significant?

```
anova_m3_m4 <- anova(m3, m4)
print(anova_m3_m4)
```

Analysis of Deviance Table

Model 1: `annual_premium ~ bmi + smoker + alcohol_freq + age + sex + education + income + marital_status + household_size`

Model 2: `annual_premium ~ bmi * smoker + alcohol_freq + age + sex + education + income + marital_status + household_size`

	Resid. Df	Resid. Dev	Df	Deviance	F	Pr(>F)
1	99980	1.5281e+10				
2	99978	1.5281e+10	2	100979	0.3303	0.7187

```
# Model selection criteria
```

```
# GLMs don't have standard r.squared. Calculating Pseudo R-squared: 1 - (deviance / null.deviance)
```

```
r2_m1 <- 1 - (m1$deviance / m1$null.deviance)
```

```
r2_m2 <- 1 - (m2$deviance / m2$null.deviance)
```

```
r2_m3 <- 1 - (m3$deviance / m3$null.deviance)
```

```
r2_m4 <- 1 - (m4$deviance / m4$null.deviance)
```

```
model_comparison <- data.frame(
```

```
  Model = c("Modifiable Only", "Baseline Only", "Full Model", "Full + Interaction"),
```

```
  AIC = c(AIC(m1), AIC(m2), AIC(m3), AIC(m4)),
```

```
  BIC = c(BIC(m1), BIC(m2), BIC(m3), BIC(m4)),
```

```
  # Use the calculated Pseudo R-squared values here
```

```
  Pseudo_R_squared = c(r2_m1, r2_m2, r2_m3, r2_m4)
```

```
)
```

```
print(model_comparison)
```

	Model	AIC	BIC	Pseudo_R_squared
1	Modifiable Only	1479233	1479309	0.02619346
2	Baseline Only	1480247	1480389	0.01641006
3	Full Model	1477528	1477727	0.04290948
4	Full + Interaction	1477531	1477750	0.04291581

Formal model comparison tests using ANOVA and information criteria consistently identified the full model (M3) incorporating both modifiable lifestyle factors and baseline demographic/socioeconomic variables as the optimal specification. The addition of modifiable factors (BMI, smoking status, alcohol frequency) to the baseline model significantly improved model fit ($F(6, 99980) = 461.37$, $p < 0.001$), while baseline factors also provided significant incremental explanatory power beyond modifiable factors alone ($F(13, 99980) = 134.32$, $p < 0.001$). However, the BMI $\times$ smoking interaction term did not significantly improve model fit ($F(2, 99978) = 0.33$, $p = 0.719$), suggesting that the relationship between BMI and premiums

does not meaningfully differ across smoking categories. Model selection criteria further support the full model without interaction as the preferred specification: the full model exhibited the lowest AIC (1,477,528) and BIC (1,477,727), with $R^2 = 0.043$ indicating that modifiable and baseline factors combined explain 4.3% of variance in insurance premiums. While the Full + Interaction model showed negligibly similar R^2 (0.043), the lack of statistical significance coupled with higher AIC/BIC values due to increased model complexity argues against its inclusion. These results demonstrate that both modifiable lifestyle choices and demographic characteristics independently contribute to insurance premium variation, justifying a comprehensive modeling approach that accounts for both factor sets.

```
# Calculate proportions of explained variance only (4.3% of total variance)
#total_explained <- rsq_full
#prop_modifiable <- variance_modifiable_unique / total_explained
#prop_baseline <- variance_baseline_unique / total_explained
#prop_shared <- variance_overlap / total_explained

#cat("Of the explained variance:\n")
#cat("Modifiable:", prop_modifiable * 100, "%\n")
#cat("Baseline:", prop_baseline * 100, "%\n")
#cat("Shared:", prop_shared * 100, "%\n")

# Create visualization
#variance_data_explained <- data.frame(
#   Component = factor(c("Modifiable\n(Unique)", "Baseline\n(Unique)", "Shared")),
#   levels = c("Modifiable\n(Unique)", "Baseline\n(Unique)", "Shared")),
#   Percentage = c(
#     prop_modifiable * 100,
#     prop_baseline * 100,
#     prop_shared * 100
#   )
#)

#ggplot(variance_data_explained, aes(x = "", y = Percentage, fill = Component)) +
#  geom_bar(stat = "identity", width = 1, color = "white", linewidth = 1) +
#  coord_polar("y", start = 0) +
#  scale_fill_manual(values = c("Modifiable\n(Unique)" = "lightblue",
#                                "Baseline\n(Unique)" = "maroon",
#                                "Shared" = "orange")) +
#  labs(title = "Variance Partitioning: Explained Variance Only",
#        subtitle = "Of the 4.3% explained variance, what proportion is modifiable vs. baseline",
#        fill = "") +
#  theme_void() +
#  theme(plot.title = element_text(face = "bold", size = 12, hjust = 0.5),
```

```
#       plot.subtitle = element_text(size = 10, hjust = 0.5, margin = margin(b = 20))) +  
#   geom_text(aes(label = sprintf("%.1f%%", Percentage)),  
#             position = position_stack(vjust = 0.5),  
#             size = 4, fontface = "bold")
```

Individual-level characteristics explained only 4.3% of total premium variance, which makes sense given that insurance pricing is heavily influenced by policy structure and market factors. To answer our research question, we looked at how this 4.3% breaks down between modifiable and baseline factors. Figure X shows that among the variance we could explain with individual characteristics, modifiable lifestyle factors accounted for 61.8% while baseline demographic factors accounted for 39.0%. The two sets of predictors showed almost no overlap (about 1%), meaning modifiable and baseline factors work mostly independently in determining premiums. This shows that when it comes to the individual characteristics that do matter for premiums, lifestyle choices are much more important than things like age or sex that people cannot change. These findings are relevant for insurers designing pricing models and for individuals who want to lower their healthcare costs by changing their lifestyle habits.